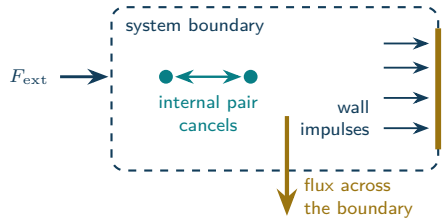
MOMENTUM ON ONE PAGE

The momentum hinge

One rate equation, $F_{\text{ext}} = \mathrm{d}p_{\text{sys}}/\mathrm{d}t$, and three mechanisms of transfer: a collision, a rocket and a gas.

1 / THE RULE AND THE PICTURE

Force is the rate at which momentum changes or crosses a chosen boundary



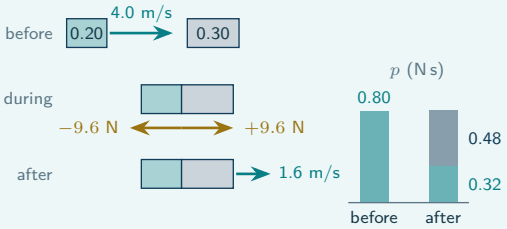
Begin with the rate equation, state the system, then add the closure that makes the force calculable.

$$F_{\text{ext}} = \frac{\mathrm{d}p_{\text{sys}}}{\mathrm{d}t}$$

Collision: an internal pair. **Rocket**: a flux across the boundary. **Gas**: wall impulses per unit area.

2 / COLLISION / IMPULSE AND INTERNAL FORCE

Equal and opposite impulses move momentum between the trolleys

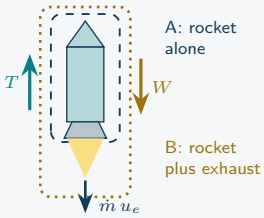


A 0.20 kg trolley at $4.0\ \mathrm{m\,s^{-1}}$ locks onto a stationary 0.30 kg trolley over $\Delta t = 0.050$ s, with negligible external impulse. Conservation, $0.20(4.0) = 0.50\,v_f$, is one equation, and locking closes it: $v_f = 1.6\ \mathrm{m\,s^{-1}}$. Each trolley has $\Delta p = \mp 0.48$ N s, so $F_{\text{av}} = \mp 9.6$ N, an equal and opposite pair, while the pair has $\Delta p = 0$ and $F_{\text{ext}} \approx 0$. A shorter Δt raises F_{av} at fixed impulse, and kinetic energy falls from 1.6 J to 0.64 J.

$$F_{\text{av}} = \Delta p/\Delta t = \pm 9.6\ \text{N on each trolley}, \qquad F_{\text{ext}} \approx 0\ \text{on the pair}$$

3 / ROCKET / MOMENTUM CROSSING A BOUNDARY

Exhaust carries momentum away, and the rocket gains the balance

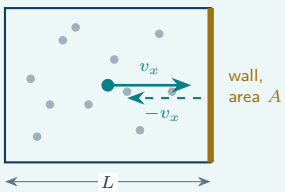


A 1200 kg rocket expels gas at $30\ \mathrm{kg\,s^{-1}}$ with relative speed $u_e = 600\ \mathrm{m\,s^{-1}}$ and $p_e = p_a$. In Δt the expelled mass $\dot{m}\,\Delta t$ carries momentum $u_e\,\dot{m}\,\Delta t$ away, so $T = \dot{m}\,u_e = 1.8 \times 10^4$ N (3600 N s in 0.20 s); with $W = 1.18 \times 10^4$ N the net force is 6240 N and $a = 5.2\ \mathrm{m\,s^{-2}}$ upward. Boundary B sees an internal redistribution, while boundary A sees mass and momentum crossing at the nozzle, so a fixed-mass equation lacks the flux term; if $p_e \neq p_a$ then $T = \dot{m}\,u_e + (p_e - p_a)A_e$.

$$T = \dot{m}\,u_e = 1.8 \times 10^4\ \text{N}, \qquad a = (T - mg)/m = 5.2\ \mathrm{m\,s^{-2}}$$

4 / GAS PRESSURE / MOLECULAR MOMENTUM FLUX

Many small impulses average into a steady pressure



A molecule bouncing elastically off a wall changes momentum by $2mv_x$ and returns after $\Delta t = 2L/|v_x|$, so $F_i = mv_x^2/L$. Summing over N molecules with $V = AL$ gives $P = Nm\langle v_x^2\rangle/V$, and isotropy, $\langle v_x^2\rangle = \frac{1}{3}\langle v^2\rangle$, gives $P = Nmc_{\text{rms}}^2/(3V) \approx 9.7 \times 10^4$ Pa for $L = 0.20$ m, $N = 2.0 \times 10^{23}$, $m = 4.65 \times 10^{-26}$ kg, $c_{\text{rms}} = 500\ \mathrm{m\,s^{-1}}$. Then $PV = \frac{2}{3}E_k$, and with $E_k = \frac{3}{2}NkT$ this is $PV = NkT$ ($T \approx 281$ K): pressure is momentum flux per unit area.

$$P = Nmc_{\text{rms}}^2/(3V) \approx 9.7 \times 10^4\ \text{Pa}, \qquad PV = \tfrac{2}{3}E_k$$

5 / SYNTHESIS

Force is a rate of momentum transfer, but the boundary decides how the transfer is seen

case	momentum system	closure or model	what produces the force
collision	both trolleys, contact pair internal	lock together, no external impulse	equal and opposite contact forces over Δt
rocket	rocket plus exhaust, or rocket alone	exhaust flux, $p_e = p_a$	momentum carried off by expelled mass, $T = \dot{m}\,u_e$
gas	molecules in the box, wall as boundary	elastic collisions, isotropy	repeated molecular impulses, P per unit area

One equation, $F_{\text{ext}} = \mathrm{d}p/\mathrm{d}t$, therefore covers contact force, thrust and pressure; the system and the closure say which.

ASIDE / THREE MISCONCEPTIONS

"Momentum is conserved for each trolley" holds only for the isolated pair. "The exhaust pushes on the air" fails because thrust persists in a vacuum. "Pressure is a stored substance" fails because it is a momentum flux per unit area.

ASIDE / CHECK THE NUMBERS IN PYTHON

`vf = 0.2*4/0.5` gives 1.6, `0.2*(vf-4)/0.05` gives -9.6, `(30*600-1200*9.8)/1200` gives 5.2, and `2e23*4.65e-26*500**2/(3*0.2**3)` gives 9.69×10^4.

Convention. After the worksheet *The Momentum Hinge*, a companion to *The Placeholder Force*; $m = 4.65 \times 10^{-26}$ kg is an N_2 molecule.

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